VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

ATTENDANCE SYSTEM

A8517-Database Management Systems Laboratory

Submittedby

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VARDHAMAN
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This is to certify that the Course End Project titled **"ATTENDANCE SYSTEM"** is carried out by **C. Ram Charan Teja**, **M. Keerthan**, and **P. Mokshith** with Roll Numbers **24881A05E2**, **24881A05G8**, and **24881A05H5** towards **A8517 – Database Management Systems Laboratory** course in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** during the Academic Year **2025–26.**

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Abstract

The Student Attendance System is a database-oriented application developed to efficiently manage and maintain student attendance records in educational institutions. Traditional methods of recording attendance manually are time-consuming, error-prone, and difficult to manage, especially when dealing with a large number of students. This system aims to automate the attendance process by providing a structured and reliable digital platform where attendance data can be stored, retrieved, and analyzed with ease. It enables faculty members to mark attendance electronically, view records, and generate reports without any redundancy or data inconsistency.

The system is designed using fundamental concepts of Database Management Systems (DBMS), including entity-relationship modeling, relational schema design, and normalization techniques. Various constraints such as primary key, foreign key, NOT NULL, UNIQUE, and CHECK are applied to ensure data integrity and consistency within the database. The implementation also focuses on maintaining ACID properties, ensuring secure and reliable transactions. By using SQL queries for data manipulation and retrieval, the system enhances performance and provides accurate results. Overall, the Student Attendance System simplifies attendance management, reduces manual workload, and contributes to efficient academic administration, making it a practical solution for modern educational environments.

Keywords: Attendance, DBMS, Student Records, SQL, Constraints, Automation, Data Integrity, ER Model

1 Introduction

Attendance tracking is an essential part of academic institutions. Traditionally, attendance is recorded manually, which is time-consuming and prone to errors. Maintaining physical records also makes it difficult to analyze attendance trends and generate reports.

The Student Attendance System is designed to automate this process using a database management system. It stores attendance records digitally and allows quick retrieval, modification, and analysis of data. This system improves efficiency, reduces redundancy, and ensures data accuracy using DBMS concepts like normalization and constraints.

In addition to improving efficiency, the Student Attendance System also enhances transparency and accountability in attendance management. Students and faculty can easily access attendance records, reducing disputes and misunderstandings. The system can be further extended to include features such as percentage calculation, alerts for low attendance, and integration with academic performance systems. By digitizing attendance, institutions can make better data-driven decisions and improve overall academic monitoring and discipline.

2 Literature Review

Existing attendance systems include manual registers and basic digital spreadsheets. Manual systems are prone to errors and difficult to maintain, while spreadsheets lack proper data validation and security.

Modern systems use database-driven applications that provide better accuracy, security, and scalability. Technologies like SQL databases and web-based interfaces are commonly used to manage attendance efficiently.

This project improves upon existing solutions by implementing proper relational database design, enforcing constraints, and ensuring data consistency using normalization techniques.

3 Objectives

- To design a database system for managing student attendance efficiently
- To reduce manual errors and improve data accuracy
- To implement constraints for maintaining data integrity

4 Methodology

4.1Architectural Diagram

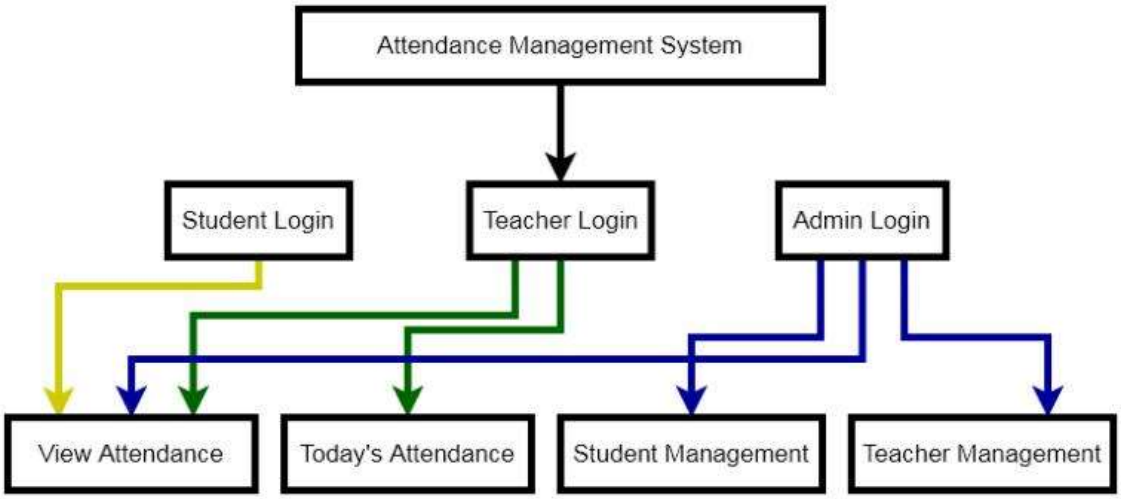
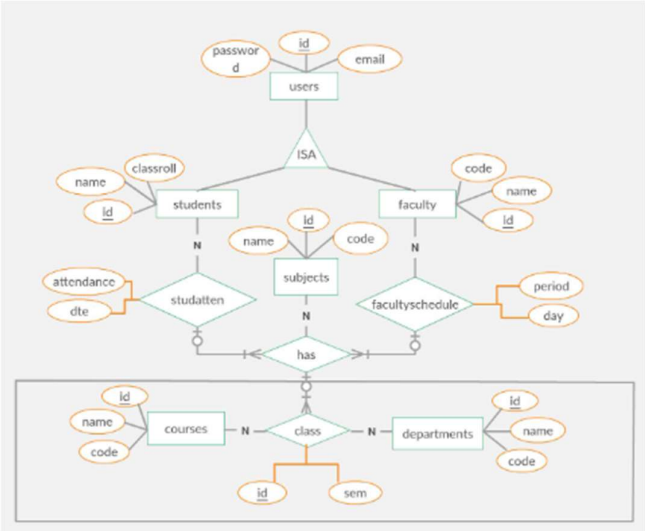


Figure 1: Architectural Diagram

4.2 Database Design



Entity Description:

- Student:This entity stores the details of students such as *SID, Name, Branch, Section*. The SID acts as primary key to uniquely identify each student in system.
- Faculty:This entity contains information about faculty members including *Faculty_ID and Name*. Faculty_ID is the primary key used to uniquely identify each faculty.

- Subject:
  This entity stores subject-related details such as *Subject_ID ,Subject_Name*. It include Faculty_ID as a foreign key to indicate which faculty teaches the subject.

- Attendance:
  This entity records the attendance details of students. It includes attributes like *Attendance_ID, Student_ID, Subject_ID, Date,* and *Status (Present/Absent)*. This entity acts as a link between Student and Subject.

Relationships:

- A Student attends Subject relationship exists where students attend multiple subjects during a semester.

- A Faculty teaches Subject relationship defines that each subject is handled by a faculty member.

- The Attendance entity connects Student and Subject, recording whether a student is present or absent for a particular subject on a given date.

Cardinality:

- One Student can have many Attendance records → *(1 : M)*

- One Subject can have many Students → *(1 : M)*

- One Faculty can teach multiple Subjects → *(1 : M)*

- Each Attendance record belongs to one Student and one Subject → *(M : 1)*

4.3 Relational Model

Schema Definition: The relational model for the Student Attendance System is designed based on the entities identified in the ER diagram. Each entity is converted into a table with appropriate primary keys and foreign keys to maintain relationships between them.

Relational Model: Tables

Student Table

Student_ID (PK)	Name	Branch	Section
101	Ramesh	CSE	A
102	Suresh	CSE	B
103	Anitha	ECE	A

Faculty Table

Faculty_ID (PK)	Name
201	Dr. Rao
202	Mrs. Sharma

Subject Table

Subject_ID (PK)	Subject_Name	Faculty_ID (FK)
301	DBMS	201
302	OS	202

Attendance Table

Attendance_ID (PK)	Student_ID (PK)	Subject_ID (PK)	Date	Status
1	101	301	2026-04-01	Present
2	102	301	2026-04-01	Absent
3	103	302	2026-04-01	Present

Relational Schema Representation:

- Student(Student_ID, Name, Branch, Section)
- Faculty(Faculty_ID, Name)
- Subject(Subject_ID, Subject_Name, Faculty_ID)
- Attendance(Attendance_ID, Student_ID, Subject_ID, Date, Status)

4.4 Normalization

1. 1NF: All attributes contain atomic (single) values **,** No repeating groups or multivalued attributes exist.So all tables are in 1NF.

2. 2NF: It is in 1NF **,**All non-key attributes are fully dependent on the primary key.Therefore all tables are in 2NF.

3.3NF: It is in 2NF,There is no transitive dependency (non-key attributes should not depend on other non-key attributes).Therefore,All tables are in 3NF.

Thus, the database schema is well normalized up to 3NF, ensuring minimal redundancy and efficient data organization.

4.5 Hardware and Software Requirements

HARDWARE:

Computer or Laptop of Minimum 4 GB RAM ,Minimum Intel i3 processor or equivalent ,Hard disk with at least 10 GB free space.

SOFTWARE:

Operating System: Windows / Linux / macOS

Database: MySQL / Oracle / PostgreSQL

4.6 Working Principle

The Student Attendance System operates based on a structured database approach where all student, faculty, subject, and attendance data are stored in relational tables. The system allows authorized users, such as faculty members, to interact with the application through a user interface.

The system processes user inputs through the application layer, where validation checks are performed to ensure accuracy and consistency of data. Constraints such as primary keys, foreign keys, and check conditions are enforced to maintain data integrity

Furthermore, the system enables retrieval and analysis of attendance data through queries and reports. The overall working principle ensures a reliable, efficient, and error-free mechanism for managing attendance, reducing manual effort and improving accuracy in academic record keeping.

4.7 Database Implementation

DDL:

```
CREATE TABLE Student (
  Student_ID   INT PRIMARY KEY,
  Name       VARCHAR(50) NOT NULL,
  Branch     VARCHAR(50),
  Section    VARCHAR(10)
);
CREATE TABLE Faculty (
  Faculty_ID   INT PRIMARY KEY,
  Name       VARCHAR(50) NOT NULL
);
CREATE TABLE Subject (
  Subject_ID    INT PRIMARY KEY,
  Subject_Name  VARCHAR(50),
  Faculty_ID    INT,
  FOREIGN KEY (Faculty_ID) REFERENCES Faculty(Faculty_ID)
);
CREATE TABLE Attendance (
  Attendance_ID INT PRIMARY KEY,
  Student_ID    INT,
  Subject_ID    INT,
  Date        DATE,
  Status      VARCHAR(10) CHECK (Status IN ('Present','Absent')),
  FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),
  FOREIGN KEY (Subject_ID) REFERENCES Subject(Subject_ID)
);
```

DML:

```
INSERT INTO Faculty VALUES (201, 'Dr. Rao');

INSERT INTO Subject VALUES (301, 'DBMS', 201);

INSERT INTO Student VALUES (101, 'Ramesh', 'CSE', 'A');

INSERT INTO Attendance VALUES (1, 101, 301, '2026-04-01', 'Present');

UPDATE Attendance

SET Status = 'Absent'

WHERE Attendance_ID = 1;

DELETE FROM Attendance

WHERE Attendance_ID = 1;
```

JOIN :

```
SELECT S.Name, Sub.Subject_Name, A.Date, A.Status

FROM Student S

JOIN Attendance A ON S.Student_ID = A.Student_ID

JOIN Subject Sub ON A.Subject_ID = Sub.Subject_ID;
```

AGGREGATE FUNCTIONS:

```
SELECT COUNT(*) FROM Attendance;

SELECT COUNT(*)

FROM Attendance

WHERE Status = 'Present'

SELECT Student_ID, COUNT(*) AS Total_Attendance

FROM Attendance

GROUP BY Student_ID;
```

Additional Constraints:

```
ALTER TABLE AttendanceMODIFY Status VARCHAR(10) DEFAULT 'Absent';

ALTER TABLE Student

ADD UNIQUE(Name);
```

4.8 Advanced DBMS concepts

ACID properties ensure that database transactions are processed reliably maintain data integrity.

- Atomicity:Each transaction is completed fully or not executed at all. For example, while marking attendance, either the entire record is inserted successfully or no change is made.
- Consistency:The database remains in a valid state before and after a transaction. Constraints such as primary key, foreign key conditions ensure only valid data is stored.
- Isolation:Multiple users can access the database simultaneously without interfering with each other. For instance, different faculty members can mark attendance independently.
- Durability:Once a transaction is committed, the data is permanently stored in the database, even in case of system failure.

Concurrency control manages simultaneous access to the database to prevent conflicts and maintain consistency.

- The system allows multiple users (faculty/admin) to access and update attendance records at the same time.
- Techniques such as locking mechanisms are used to avoid data inconsistency.
- Transactions are executed in a controlled manner to prevent issues like lost updates or dirty reads.

Indexing improves the speed of data retrieval operations.

- Indexes are created on frequently accessed columns such as Student_ID and Subject_ID.
- This reduces the time required to search and retrieve attendance records.
- It enhances overall system performance, especially when handling large datasets.

4.9 Software Implementation

Step 1: Start

Step 2: Initialize the system and establish connection with the database.

Step 3: Display login interface to the user (Faculty/Admin).

Step 4: Enter username and password.

Step 5: Validate user credentials:

If valid → proceed to next step

Else → display error message and return to Step 3

Step 6: Display main menu options:

1. Add Student
2. Add Subject
3. Mark Attendance
4. View Attendance
5. Exit

Step 7: If user selects "Add Student":

- Enter student details (ID, Name, Branch, Section) Insert data into Student table

Step 8: If user selects "Add Subject":

- Enter subject details (ID, Name, Faculty_ID Insert data into Subject table

Step 9: If user selects "Mark Attendance":

- Select Student_ID and Subject_ID ,Enter date and attendance status (Present/Absent) and Store record in Attendance table

Step 10: If user selects "View Attendance":

- Retrieve attendance records from database Display attendance

Step 11: Perform validation checks:

- Ensure no duplicate entries Ensure valid foreign key reference Ensure status is either Present or Absent

Step 12: Repeat steps based on user choice until Exit is selected.

Step 13: Close database connection.

Step 14: End

5 ResultsandDiscussion

```
SQL> SELECT * FROM Student;

STUDENT_ID NAME                                               BRANCH                                            SECTION
---------- -------------------------------------------------- -------------------------------------------------- ----------
        22 Ramcharan                                          CSE                                               C
        13 Mokshith                                           CSE                                               C
        78 Keerthan                                           CSE                                               C
        45 Rahul                                              CSE                                               C

SQL> SELECT * FROM Subject;

SUBJECT_ID SUBJECT_NAME                                       FACULTY_ID
---------- -------------------------------------------------- ----------
       301 DBMS                                                     201
       302 COSM                                                     202
       303 DAA                                                      203

SQL> SELECT * FROM Attendance;

ATTENDANCE_ID STUDENT_ID SUBJECT_ID ATTENDANC STATUS
------------- ---------- ---------- --------- ----------
            1         22        301 01-APR-26 Present
            2         13        301 01-APR-26 Absent
            3         78        301 01-APR-26 Present
            4         45        301 01-APR-26 Present
            5         22        302 02-APR-26 Present
            6         13        302 02-APR-26 Present
            7         78        302 02-APR-26 Absent
            8         45        302 02-APR-26 Present
            9         22        303 03-APR-26 Absent
           10         13        303 03-APR-26 Present
           11         78        303 03-APR-26 Present
           12         45        303 03-APR-26 Absent

12 rows selected.
```

```
SQL> SELECT S.Name, Sub.Subject_Name, F.Name AS Faculty_Name,
  2         A.Attendance_Date, A.Status
  3  FROM Student S
  4  JOIN Attendance A ON S.Student_ID = A.Student_ID
  5  JOIN Subject Sub ON A.Subject_ID = Sub.Subject_ID
  6  JOIN Faculty F ON Sub.Faculty_ID = F.Faculty_ID;

NAME                                               SUBJECT_NAME                                       FACULTY_NAME                                       ATTENDANC STATU
S
-------------------------------------------------- -------------------------------------------------- -------------------------------------------------- --------- -----
-----
Ramcharan                                          DAA                                                Vijay                                              03-APR-26 Absen
t
Ramcharan                                          COSM                                               Chandrashekar                                      02-APR-26 Prese
nt
Ramcharan                                          DBMS                                               Varma                                              01-APR-26 Prese
nt
Mokshith                                           DAA                                                Vijay                                              03-APR-26 Prese
nt
Mokshith                                           COSM                                               Chandrashekar                                      02-APR-26 Prese
nt
Mokshith                                           DBMS                                               Varma                                              01-APR-26 Absen
t
Keerthan                                           DAA                                                Vijay                                              03-APR-26 Prese
nt
Keerthan                                           COSM                                               Chandrashekar                                      02-APR-26 Absen
t
Keerthan                                           DBMS                                               Varma                                              01-APR-26 Prese
nt
Rahul                                              DAA                                                Vijay                                              03-APR-26 Absen
t
Rahul                                              COSM                                               Chandrashekar                                      02-APR-26 Prese
nt
Rahul                                              DBMS                                               Varma                                              01-APR-26 Prese
nt

12 rows selected.
```

```
SQL> SELECT S.Name, Sub.Subject_Name, F.Name AS Faculty_Name, A.Status
  2  FROM Student S
  3  JOIN Attendance A ON S.Student_ID = A.Student_ID
  4  JOIN Subject Sub ON A.Subject_ID = Sub.Subject_ID
  5  JOIN Faculty F ON Sub.Faculty_ID = F.Faculty_ID;

NAME                                               SUBJECT_NAME                                       FACULTY_NAME
                                         STATUS
-------------------------------------------------- -------------------------------------------------- --------------------------
Ramcharan                                          DAA                                                Vijay
                                         Absent
Ramcharan                                          COSM                                               Chandrashekar
                                         Present
Ramcharan                                          DBMS                                               Varma
                                         Present
Mokshith                                           DAA                                                Vijay
                                         Present
Mokshith                                           COSM                                               Chandrashekar
                                         Present
Mokshith                                           DBMS                                               Varma
                                         Absent
Keerthan                                           DAA                                                Vijay
                                         Present
Keerthan                                           COSM                                               Chandrashekar
                                         Absent
Keerthan                                           DBMS                                               Varma
                                         Present
Rahul                                              DAA                                                Vijay
                                         Absent
Rahul                                              COSM                                               Chandrashekar
                                         Present
Rahul                                              DBMS                                               Varma
                                         Present

12 rows selected.
```

Run SQL Command Line

```
SQL> SELECT COUNT(*) AS Total_Records FROM Attendance;

TOTAL_RECORDS
-------------
           12

SQL> SELECT S.Name, COUNT(*) AS Total_Classes
  2  FROM Student S
  3  JOIN Attendance A ON S.Student_ID = A.Student_ID
  4  GROUP BY S.Name;

NAME                                               TOTAL_CLASSES
-------------------------------------------------- -------------
Ramcharan                                                      3
Rahul                                                          3
Keerthan                                                       3
Mokshith                                                       3

SQL> SELECT S.Name, COUNT(*) AS Present_Count
  2  FROM Student S
  3  JOIN Attendance A ON S.Student_ID = A.Student_ID
  4  WHERE A.Status = 'Present'
  5  GROUP BY S.Name;

NAME                                               PRESENT_COUNT
-------------------------------------------------- -------------
Ramcharan                                                      2
Rahul                                                          2
Keerthan                                                       2
Mokshith                                                       2
```

6 Mapping POsand SDGs

6.3 Program Outcomes (POs)Mapping

Table1: Mapping of Program Outcomes (POs) to Project Relevance

PONo.	ProgramOutcome	Relevance
PO1	EngineeringKnowledge	Applied DBMS concepts such as relational models, normalization, and SQL queries in developing the attendance system.
PO2	ProblemAnalysis	Identified issues in manual attendance systems and analyzed requirements to design an efficient digital solution.
PO3	Design/DevelopmentofSolutions	Designed &implemented a database system for managing student attendance effectively.
PO4	Conduct Investigations of Complex Problems	Tested queries and analyzed outputs to ensure correctness ,efficiency of the system.
PO5	Engineering Tool Usage	Used tools like Oracle/MySQL and IDEs to implement and test the database system.
PO6	The Engineer and The World	Developed a system useful for institutions, improving academic management.
PO7	Ethics	Ensured accurate and secure handling of student data without manipulation or misuse.
PO8	Individual and Collaborative Team work	Worked individually/in a team to design, implement, and test the system.
PO9	Communication	Documented the project clearly and presented results using reports and queries.
PO10	Project Management and Finance	Managed time and resources efficiently during development of the project.
PO11	Life-Long Learning	Gained practical knowledge of DBMS concepts and tools useful for future learning.

6.3 Sustainable Development Goals(SDGs)Mapping

Table2:Mapping of Sustainable Development Goals (SDGs)to Project Relevance

SDGNo.	Goal	Relevance
SDG4	Quality Education	The system improves academic monitoring by maintaining accurate attendance records, helping institutions ensure better learning outcomes.
SDG9	Industry,Innovation and Infrastructure	The project uses database technology to develop an efficient digital solution, promoting innovation in educational systems.
SDG10	Reduced Inequalities	Provides equal and transparent attendance tracking for all students, avoiding bias or manual errors.
SDG16	Peace, Justice and Strong Institutions	Ensures transparency and accountability in attendance records, supporting fair academic evaluation.

7 Conclusion

The Student Attendance System was successfully designed and implemented using fundamental concepts of Database Management Systems such as entity-relationship modeling, relational schema design, normalization, and SQL queries. The system replaces the traditional manual attendance process with a digital solution that ensures accuracy, efficiency, and reliability. By organizing data into structured tables such as Student, Faculty, Subject, and Attendance, the system minimizes redundancy and maintains consistency. The use of constraints like primary key, foreign key, NOT NULL, and CHECK ensures data integrity and prevents invalid entries. Additionally, various SQL operations including insertion, retrieval, join, and aggregate functions enable effective management and analysis of attendance records.

The implementation of this system demonstrates how database technologies can be effectively used to solve real-world problems in educational institutions. Faculty members can easily record attendance, while administrators can retrieve and analyze data quickly. The system reduces manual workload, eliminates errors, and improves transparency in attendance tracking. The use of advanced DBMS concepts such as ACID properties, concurrency control, and indexing further enhances the reliability and performance of the system. Overall, the project achieves its objective of developing a simple, efficient, and user-friendly attendance management system.

In future, the system can be extended by integrating modern technologies to enhance its functionality and usability. Features such as biometric or RFID-based attendance systems can be implemented to automate attendance marking and eliminate proxy attendance. A web or mobile application interface can be developed to allow real-time access to attendance data for students and faculty. Additionally, advanced analytics such as attendance percentage calculation, performance tracking, and automated alerts for low attendance can be included. Integration with other academic systems like result management can further improve efficiency. These enhancements will make the system more scalable, secure, and suitable for large-scale real-world applications.

Furthermore, the project provides a strong foundation for understanding practical database applications and enhances problem-solving skills in real-world scenarios. It demonstrates how theoretical concepts of DBMS can be effectively applied to design and implement a functional system. The experience gained through this project helps in building technical competence and prepares for future development of more complex systems. With further improvements and integration of modern technologies, the Student Attendance System can evolve into a comprehensive academic management solution widely applicable in educationalinstitutions.

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